ASSIGNMENT 02

COURSE: **Computer Organization and Assembly Language** **SECTION:** **BSSEF23-MORNING**

INSTRUCTOR: **Sir Abdul Khaliq** **DEADLINE:** **12-05-2025(Till 11:59 P.M)**

(TASK 1) Write a program that prompts the user to enter a string of decimal digits, ending with a carriage return, and prints their sum in hex on the next line. If the user enters· an illegal character, he or she should be prompted to begin again.

Sample execution:

ENTER A DECIMAL DIGIT STRING: 1299843
THE SUM OF THE DIGITS IN HEX IS 0024 **(15 POINTS)**

(TASK 2) Write a program that prompts user to enter a character input and check it its 8-bit ASCII character is Palindrome or not. Print "Palindrome" or "Not Palindrome" in next line also print the Binary value of character input.

NOTE: Palindrome number is sequence of bits that reads same backward as forward.

Sample Execution:

ENTER A CHARACTER: A
ASCII BINARY: 01000001
RESULT: PALINDROME. **(20 POINTS)**

(TASK 3) Write a program that prompts the user to enter two binary numbers of up to 8 digits each and prints their sum on the next line in binary. If the user enters an illegal character, he or she should be prompted to begin again. Each input ends with a carriage return.

Sample execution:

TYPE A BINARY NUMBER, UP TO 8 DIGITS: 11001010
TYPE A BINARY NUMBER, UP TO 8 DIGITS: 10011100
THE BINARY SUM IS 101100110. **(15 POINTS)**

NOTE:

Put the .asm files containing the code and screenshot of the output from the emulator screen in a folder named as BSSEF23M---_ASSIGNMENT2 (--- should be replaced by your roll number). You have to submit this zipped folder.